

6.1 Text Classification Pipeline

The full path from raw text to a label.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Classification is supervised

Supervised learning means the training rows come with the answer. Those answers are **labels**.

Two jobs dominate:

- **Classification:** a discrete bucket. Spam vs ham is the running example. The label is a class.
- **Regression:** a number. Price of a car from mileage and brand. The label is continuous.

This week is classification of **text**. The input is a document or snippet. The output is a predefined category.

Algorithms you already know from a first ML course still apply once the text is a vector:

Algorithm	Typical role in this course
(k)-NN	Simple baseline, no training of weights
Logistic regression	Strong linear baseline on TF-IDF
Linear SVM	Same, often a bit better on high-dimensional text
Trees / forests	Nonlinear, weaker on raw sparse bags
Neural nets	Weeks 7–8 once you have embeddings

Unsupervised methods (clustering, PCA, t-SNE, LDA) have no labels. They organize or compress. They are not this week's classifier, though you may use them to **look** at the data.

The main text is *Practical Natural Language Processing*.

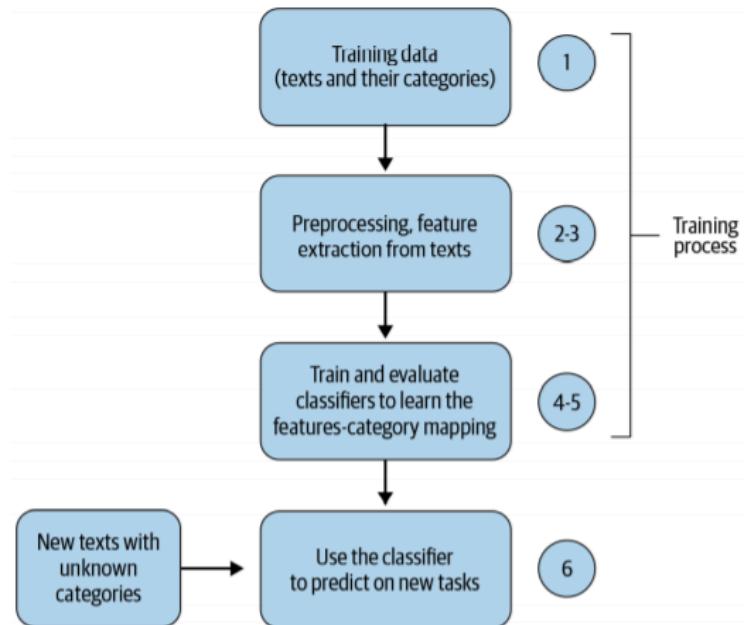
2. What text classification is

You assign each document a label from a fixed set: sentiment, topic, spam, language, priority. The model learns patterns from a **labeled** collection, then labels new text.

It is the most common applied NLP job in this course. If you can take a CSV of texts and labels and ship a scored model, you can do the rest of the semester's projects.

3. The pipeline

Every system has the same spine. Representation is Week 4–5. This week you put a **classifier** and an **evaluation** on the end.



In order:

1. **Collect and label.** The bottleneck is usually labels, not models.
2. **Clean and tokenize.** Lowercase, strip junk, decide what a token is.
3. **Vectorize.** BoW, TF-IDF, or embeddings. Fit the vectorizer on **train only**.
4. **Train.** Pick a model family and its hyperparameters on a validation split or inner CV (note **6.2**).
5. **Evaluate.** Accuracy if classes are balanced; precision, recall, F1 if they are not.
6. **Ship and watch.** New topics and new slang shift the distribution. A model that was fine in March can fail in September.

Skip a box and you will debug the wrong thing. A “bad SVM” is often a leak (vectorizer fit on the whole corpus) or a metric that hides a majority class.

4. Where this shows up

Application	What the label is
Content organization	Topic or desk (sports, politics)
Misinformation	Claim is supported / not, or source is reliable / not
Authorship	Which writer (or which account)
Customer support	Intent, queue, or urgency
E-commerce	Category, aspect sentiment, fake-review flag



Support and e-commerce are the pictures in lecture because the **cost of a wrong bucket** is obvious: the ticket goes to the wrong team, or the product lands in the wrong aisle. Build the pipeline so you can measure that cost, not just accuracy.

5. What note 6.2 adds

This note is the map. Note **6.2** is the classical models and the evaluation you will use in lab: (k)-NN, SVM, cross-validation, parameters vs hyperparameters, precision and recall.

Do not start Week 7 until you can draw the flowchart and name a metric for an imbalanced ticket queue.

6. Practice

1. Is topic modeling (LDA) text classification? Why or why not?
2. You fit `TfidfVectorizer` on train+test together, then split. What did you leak?

3. Pick one application in the table. Name the classes, who labels the data, and what a false positive costs.